

LECTURE 5

Sequence Alignment Concepts

Bioinformatics · MS Data Science

Shahid Manzoor

Global vs. local · Substitution matrices · Gap penalties · Reading a score

G	A	T	T	A	C	A
G	A	T	-	A	C	A

$$\text{score} = 6 \times (+1) + 1 \times (-2) = 4$$

One gap, six matches: is this the best alignment?

Learning objectives



Distinguish global vs. local alignment

Know what each one optimizes and choose the right one for a biological question.



Explain substitution matrices and gap penalties

PAM vs. BLOSUM, linear vs. affine gaps, and why they shape the final alignment.



Interpret an alignment score

Raw score, % identity, similarity: what they tell you about homology, and what they do not.

Why align sequences?



Infer homology

Shared ancestry explains shared sequence.



Transfer function

Annotate an unknown protein using a characterized relative.



Reconstruct evolution

Alignments are the input to phylogenetic trees.



Find variants

Align reads to a reference to spot SNPs and indels.

Similarity $\neq$ Homology

Similarity: a measurement, e.g. 42% of positions match.

Homology: a conclusion, shared ancestry. Sequences either are homologous or they are not.

Example: human α - and β -globin share roughly 40–45% identity, enough to infer homology. Both adopt the same globin fold.

Hand-align before any software

Activity 1 · 8 min

Sequence A

G	A	T	T	A	C	A
---	---	---	---	---	---	---

Sequence B

G	A	T	A	C	A
---	---	---	---	---	---

match +1

mismatch -1

gap -2

- 1 Align A and B by eye. Insert gaps (–) where you need them.
- 2 Score your alignment with the values above.
- 3 Compare with a neighbour. Is your alignment the only best one?
- 4 Bonus: what changes if a gap costs only -0.5?

Worksheet

A

--	--	--	--	--	--	--	--	--

B

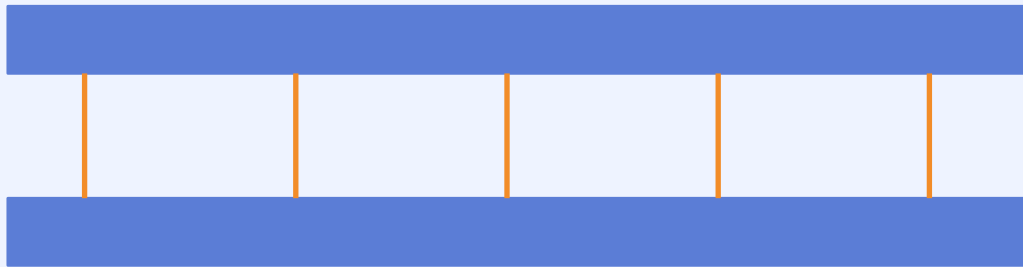
--	--	--	--	--	--	--	--	--

Score = _____ Gaps used = _____

No software yet! We will check your answer against Biopython in the lab.

Global vs. local alignment

Global · Needleman–Wunsch (1970)



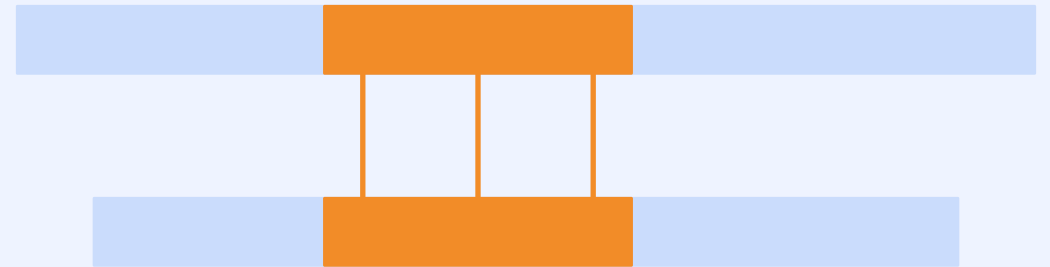
Goal: align every residue, end to end.

Best for: sequences of similar length related along their whole span (e.g., orthologs).

Scoring: can be negative; end gaps are penalized.

Watch out: forces unrelated regions into an alignment.

Local · Smith–Waterman (1981)



Goal: find the best-scoring matching region(s).

Best for: conserved domains or motifs; a short query vs. a long target.

Scoring: never below 0; the score resets to 0 outside the region.

Watch out: ignores weaker similarity outside the best region.

Scoring an alignment: worked example

Seq 1	A	C	G	T	A	G	C	+5	5 matches
Seq 2	A	C	T	T	A	-	C	-1	1 mismatch
Score	+1	+1	-1	+1	+1	-2	+1	-2	1 gap

Alignment score = Σ substitution scores (aligned pairs) + Σ gap penalties

= $5(+1) + 1(-1) + 1(-2) = 2$ *Higher is better; the algorithm finds the arrangement with the maximum score.*

Substitution matrices

Not all substitutions are equal. Leu $\leftrightarrow$ Ile is common and usually harmless; Trp $\leftrightarrow$ Asp is rare and disruptive.

DNA: 4×4. Transitions (A $\leftrightarrow$ G, C $\leftrightarrow$ T) are more frequent than transversions, so they cost less.

Protein: 20×20 log-odds scores. Positive = seen more often than by chance; negative = less often.

Try reading it: I $\leftrightarrow$ V = +3 · L $\leftrightarrow$ I = +2 · K $\leftrightarrow$ E = +1 · W $\leftrightarrow$ D = -4 · W $\leftrightarrow$ W = +11 (a rare residue that is conserved is very informative).

	L	I	V	F	W	D	E	K
L	4	2	1	0	-2	-4	-3	-2
I	2	4	3	0	-3	-3	-3	-3
V	1	3	4	-1	-3	-3	-2	-2
F	0	0	-1	6	1	-3	-3	-3
W	-2	-3	-3	1	11	-4	-3	-3
D	-4	-3	-3	-3	-4	6	2	-1
E	-3	-3	-2	-3	-3	2	5	1
K	-2	-3	-2	-3	-3	-1	1	5

BLOSUM62 excerpt · orange = favoured, blue = disfavoured

PAM vs. BLOSUM

PAM · Dayhoff et al., 1978

- Built from alignments of very close relatives ($\geq 85\%$ identical)
- PAM1 = 1 accepted point mutation per 100 residues
- Higher PAMs extrapolated by matrix multiplication (PAM250 $\approx$ distant)
- Explicit evolutionary model

BLOSUM · Henikoff & Henikoff, 1992

- Counted directly in conserved, ungapped blocks (BLOCKS database)
- BLOSUM62 clusters sequences $\geq 62\%$ identical before counting
- Lower number = more divergent sequences
- BLOSUM62: default in BLAST and most tools

Closer relatives → more distant relatives

BLOSUM80

$\approx$ PAM120 (rough match)

BLOSUM62

$\approx$ PAM160 (rough match)

BLOSUM45

$\approx$ PAM250 (rough match)

Same pair, different matrices

Activity 2 · 8 min

Peptide 1

L	K	D	W
---	---	---	---

(ungapped, position by position)

Peptide 2

I	E	D	F
---	---	---	---

Pair	Identity (+1/0)	BLOSUM62	PAM250
L / I	0	2	2
K / E	0	1	0
D / D	1	6	4
W / F	0	1	0
TOTAL	_____	_____	_____

Discuss with your neighbour

- 1 Why do L/I and K/E score 0 under identity but not under BLOSUM62?
- 2 Can you compare a BLOSUM62 total with a PAM250 total directly?
- 3 Which matrix would you choose for two distantly related proteins?

Gap penalties: linear vs. affine

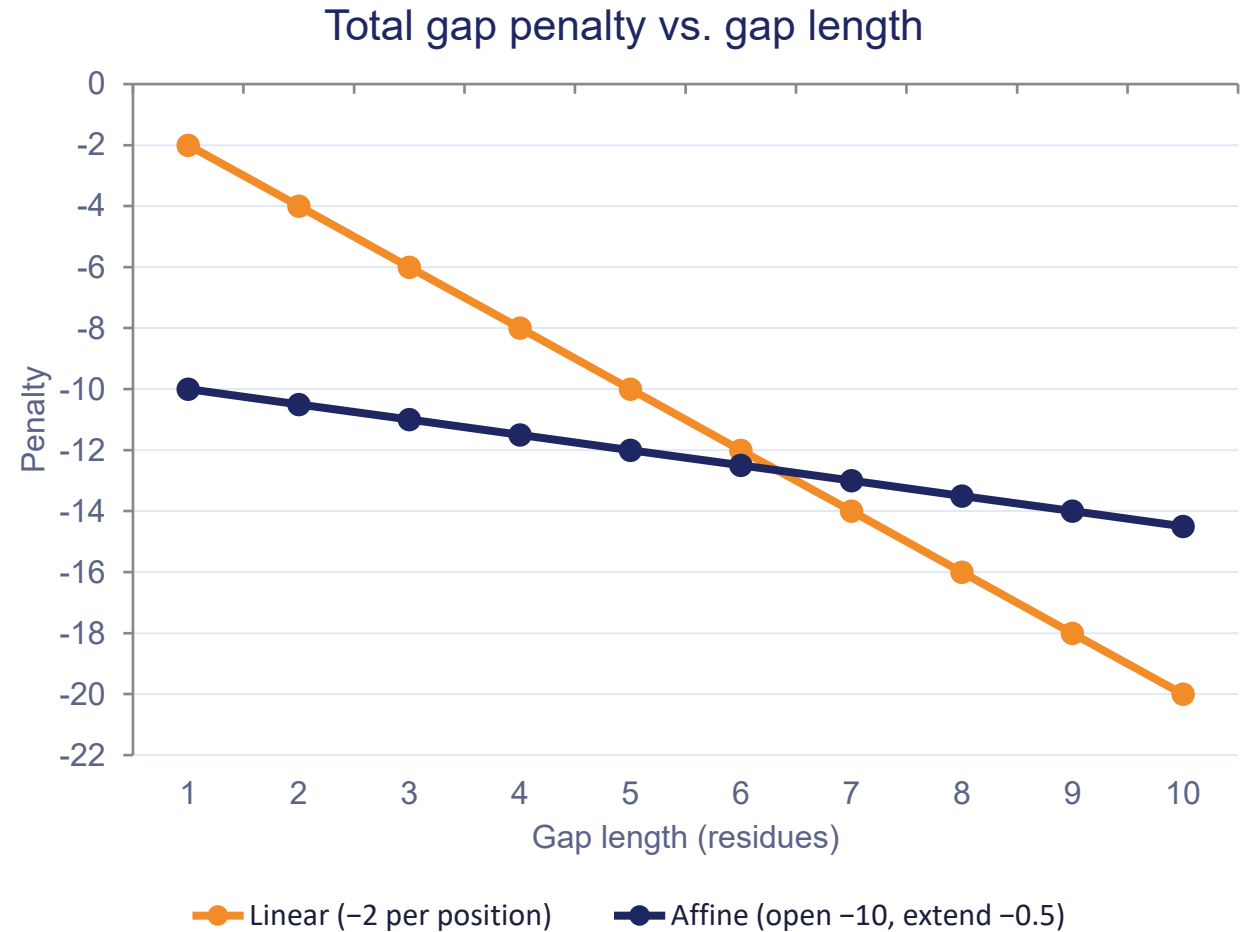
Linear: $\text{penalty} = g \times L$

e.g. -2 per position; a 5-residue gap = -10

Simple, but treats 5 separate gaps like 1 long gap.

Affine: $\text{penalty} = \text{open} + \text{extend} \times (L-1)$

e.g. open -10, extend -0.5. Insertions and deletions in real genes are often several residues long, so one long gap is more plausible than many short ones.



Why affine gaps behave better

A • one gap of length 3

X	T	A	C	G	G	A	T	C
Y	T	A	C	—	—	—	T	C

Linear (−2 each): gaps −6 → score −1

Affine (−10, −0.5): gaps −11 → score −6

B • three gaps of length 1

X	T	A	C	G	G	A	T	C
Y	T	—	C	—	G	—	T	C

Linear (−2 each): gaps −6 → score −1

Affine (−10, −0.5): gaps −30 → score −25

Same 5 matches, same 3 gapped positions. A linear penalty cannot tell A from B. An affine penalty strongly prefers A, matching biology: one insertion or deletion event often adds or removes several residues.

Interpreting an alignment score

Raw score

Sum of substitution and gap scores. Depends on matrix, gap penalties and length, so never compare across setups.

% identity / % similarity

Identity: exact matches. Similarity: matches plus conservative substitutions (positive matrix scores). Always quote the alignment length.

Bit score & E-value (BLAST)

Bit score rescales raw scores so different matrices can be compared. E-value = number of equally good hits expected by chance; smaller is more significant.

Always report: matrix · gap penalties · length · identity

Rule-of-thumb identity zones (proteins)

> ~40% over a long alignment
Homology very likely

~20–35% · twilight zone
Ambiguous: needs statistics

< ~20% · midnight zone
Use profiles or structure

Caveat: short alignments can reach high % identity by chance.

Global or local?

Quick check

1

Two orthologous enzymes of similar length from human and mouse: how similar overall?

GLOBAL

LOCAL

2

A conserved kinase domain inside two large, otherwise unrelated multi-domain proteins.

GLOBAL

LOCAL

3

Where does a 25-nt primer best match within a 5 kb gene region?

GLOBAL

LOCAL

4

Two ~1,500 bp 16S rRNA genes from the same bacterial genus: overall divergence?

GLOBAL

LOCAL

5

Does a newly sequenced protein contain a known short motif anywhere in it?

GLOBAL

LOCAL

Lab: pairwise alignment in Biopython

Practical / Lab

```
from Bio import Align
from Bio.Align import substitution_matrices

aligner = Align.PairwiseAligner()
aligner.mode = "global"      # or "local"
aligner.substitution_matrix = \
    substitution_matrices.load("BLOSUM62")
aligner.open_gap_score = -10
aligner.extend_gap_score = -0.5

alns = aligner.align("HEAGAWGHEE", "PAWHEAE")
print(alns.score, len(alns))
print(alns[0])
```

Your tasks (~30 min)

- 1 Run global, then local. What changes?
- 2 Swap in PAM250 or BLOSUM45.
- 3 Vary open/extend gaps; watch the score and len(alns).
- 4 Verify your hand-scores from both activities.

Legacy note: Bio.pairwise2 is deprecated. In older tutorials you will see pairwise2.align.globalds(a, b, blosum62, -10, -0.5).

Assignment & key takeaways



Assignment

Align 3 sequence pairs under different scoring schemes and compare the results.

- Suggested pairs: close relatives · distant homologs · a shared domain in otherwise different proteins.
- Schemes: (A) +1/-1/-2 simple · (B) BLOSUM62 with affine gaps · (C) PAM250 or BLOSUM45 with other gap values.
- Submit: a table of scores, % identity and gaps, plus one paragraph on how the scoring scheme changed each alignment.

Key takeaways

- 1 Global aligns end to end; local finds the best matching region.
- 2 Matrices score how surprising each residue pair is; pick by how diverged the sequences are.
- 3 Affine gaps: expensive to open, cheap to extend.
- 4 A score only means something together with its matrix, gap penalties and length.